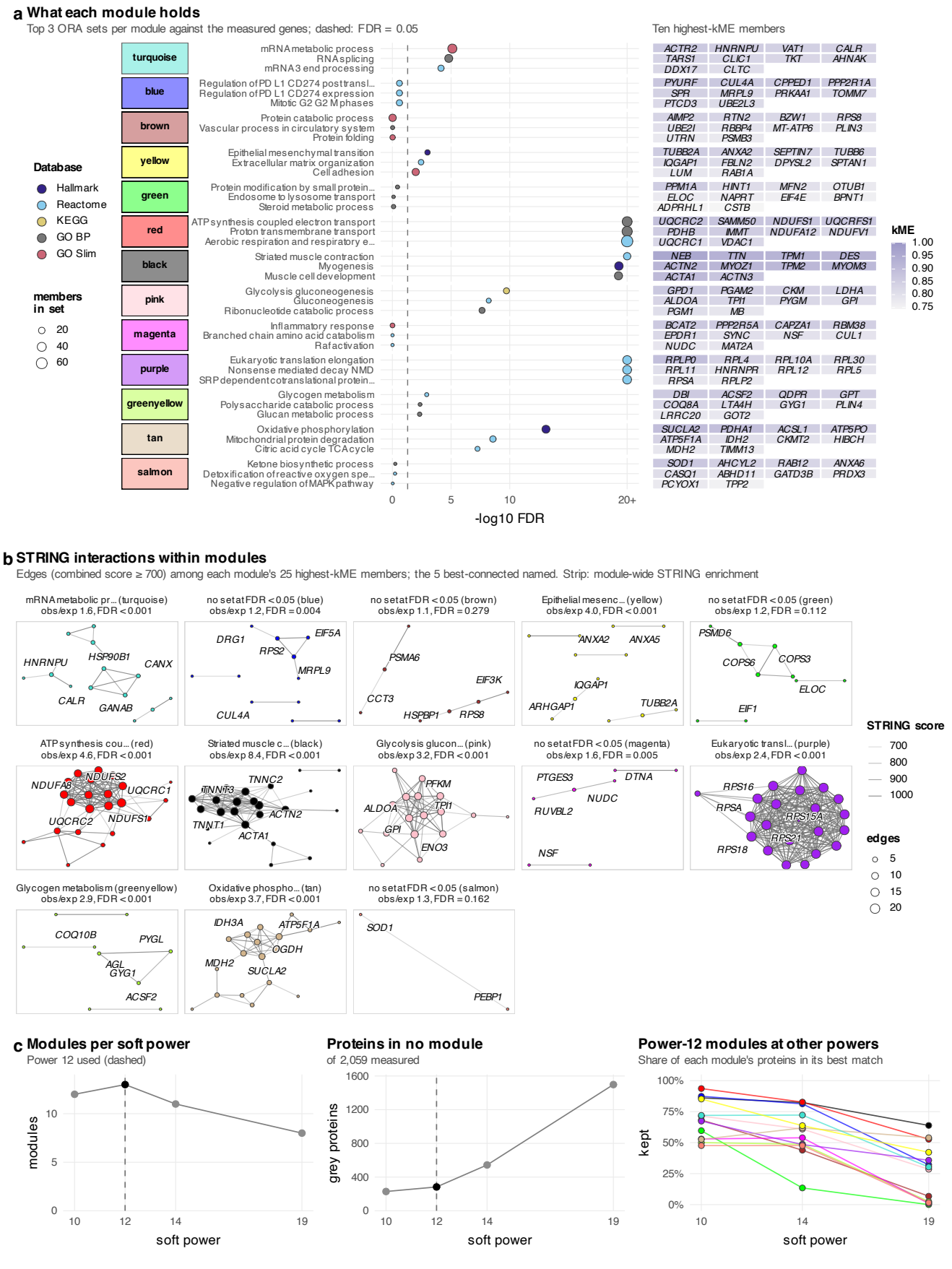

a What each module holds
Top 3 ORA sets per module against the measured genes; dashed: FDR = 0.05
Ten highest-kME members
turquoise
blue
brown
yellow
green
red
black
pink
magenta
purple
greenyellow
tan
salmon
mRNA metabolic process
RNA splicing
mRNA 3 end processing
Regulation of PD L1 CD274 posttransl...
Regulation of PD L1 CD274 expression
Mitotic G2 G2 M phases
Protein catabolic process
Vascular process in circulatory system
Protein folding
Epithelial mesenchymal transition
Extracellular matrix organization
Cell adhesion
Protein modification by small protein...
Endosome to lysosome transport
Steroid metabolic process
ATP synthesis coupled electron transport
Proton transmembrane transport
Aerobic respiration and respiratory e...
Striated muscle contraction
Myogenesis
Muscle cell development
Glycolysis gluconeogenesis
Gluconeogenesis
Ribonucleotide catabolic process
Inflammatory response
Branched chain amino acid catabolism
Rafactivation
Eukaryotic translation elongation
Nonsense mediated decay NMD
SRP dependent cotranslational protein...
Glycogen metabolism
Polysaccharide catabolic process
Glucan metabolic process
Oxidative phosphorylation
Mitochondrial protein degradation
Citric acid cycle TCA cycle
Ketone biosynthetic process
Detoxification of reactive oxygen spe...
Negative regulation of MAPK pathway
0
5
10
20+
-log10 FDR
ACTR2 HNRNPU VAT1 CALR
TARS1 CLIC1 TKT AHNAK
DDX17 CLTC
PYURF CUL4A CPPED1 PPP2R1A
SPR MRPL9 PRKAA1 TOMM7
PTCD3 UBE2L3
AIMP2 RTN2 BZW1 RPS8
UBE2I RBBP4 MT-ATP6 PLIN3
UTRN PSMB3
TUBB2A ANXA2 SEPTIN7 TUBB6
IQGAP1 FBLN2 DPYSL2 SPTAN1
LUM RAB1A
PPM1A HINT1 MFN2 OTUB1
ELOC NAPRT EIF4E BPNT1
ADPRHL1 CSTB
UQCRC2 SAMM50 NDUFS1 UQCRFS1
PDHB IMMT NDUFA12 NDUFV1
UQCRC1 VDAC1
NEB TTN TPM1 DES
ACTN2 MYOZ1 TPM2 MYOM3
ACTA1 ACTN3
GPD1 PGAM2 CKM LDHA
ALDOA TPI1 PYGM GPI
PGM1 MB
BCAT2 PPP2R5A CAPZA1 RBM38
EPDR1 SYNC NSF CUL1
NUDC MAT2A
RPLP0 RPL4 RPL10A RPL30
RPL11 HNRNPR RPL12 RPL5
RPSA RPLP2
DBI ACSF2 QDPR GPT
COQ8A LTA4H GYG1 PLIN4
LRRC20 GOT2
SUCLA2 PDHA1 ACSL1 ATP5PO
ATP5F1A IDH2 CKMT2 HIBCH
MDH2 TIMM13
SOD1 AHCYL2 RAB12 ANXA6
CASQ1 ABHD11 GATD3B PRDX3
PCYOX1 TPP2
kME
1.00
0.95
0.90
0.85
0.80
0.75
Database
Hallmark
Reactome
KEGG
GO BP
GO Slim
members in set
20
40
60
b STRING interactions within modules
Edges (combined score ≥ 700) among each module's 25 highest-kME members; the 5 best-connected named. Strip: module-wide STRING enrichment
mRNA metabolic pr... (turquoise)
obs/exp 1.6, FDR < 0.001
HNRNPU
HSP90B1
CANX
CALR
GANAB
no set at FDR < 0.05 (blue)
obs/exp 1.2, FDR = 0.004
DRG1
EIF5A
RPS2
MRPL9
CUL4A
no set at FDR < 0.05 (brown)
obs/exp 1.1, FDR = 0.279
PSMA6
EIF3K
CCT3
HSPBP1
RPS8
Epithelial mesenc... (yellow)
obs/exp 4.0, FDR < 0.001
ANXA2
ANXA5
IQGAP1
ARHGAP1
TUBB2A
no set at FDR < 0.05 (green)
obs/exp 1.2, FDR = 0.112
PSMD6
COPS6
COPS3
ELOC
EIF1
ATP synthesis cou... (red)
obs/exp 4.6, FDR < 0.001
NDUFA8
NDUFS2
UQCRC1
UQCRC2
NDUFS1
Striated muscle c... (black)
obs/exp 8.4, FDR < 0.001
TNNC2
TNNT3
ACTN2
TNNT1
ACTA1
Glycolysis glucon... (pink)
obs/exp 3.2, FDR < 0.001
PFKM
ALDOA
TPI1
GPI
ENO3
no set at FDR < 0.05 (magenta)
obs/exp 1.6, FDR = 0.005
PTGES3
DTNA
NUDC
RUVBL2
NSF
Eukaryotic transl... (purple)
obs/exp 2.4, FDR < 0.001
RPS16
RPSA
RPS15A
RPS21
RPS18
STRING score
700
800
900
1000
edges
5
10
15
20
Glycogen metabolism (greenyellow)
obs/exp 2.9, FDR < 0.001
COQ10B
PYGL
AGL
GYG1
ACSF2
Oxidative phospho... (tan)
obs/exp 3.7, FDR < 0.001
IDH3A
ATP5F1A
OGDH
MDH2
SUCLA2
no set at FDR < 0.05 (salmon)
obs/exp 1.3, FDR = 0.162
SOD1
PEBP1
c Modules per soft power
Power 12 used (dashed)
modules
soft power
Proteins in no module
of 2,059 measured
grey proteins
soft power
Power-12 modules at other powers
Share of each module's proteins in its best match
kept
soft power